MOBILE-MONEY ANTI-MONEY LAUNDERING DECISION-SUPPORT SYSTEM

CHAPTER 1

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1.1 Introduction

Digital financial services have expanded access to payments and transfers through mobile wallets and agent networks. Mobile money has become particularly important in Sub-Saharan Africa, where it supports financial activity beyond conventional bank branches [1]. Zimbabwe's National Financial Inclusion Strategy II recognised the importance of access to, usage of and quality of financial services [2]. This context provided the motivation for examining transaction-monitoring support for mobile-money services.

This study addressed the identification of suspicious patterns that may emerge across transaction histories, relationships between wallets and agent activity. It developed a local mobile-money Anti-Money Laundering (AML) decision-support prototype combining behavioural analysis, machine-learning classification and an investigation workflow. Its classifications were intended to support human review rather than establish that money laundering had occurred. This chapter presents the background, problem, aim, objectives, limitations, delimitations, development instruments, work plan and justification.

1.2 Background of the Study

International AML monitoring has increasingly considered how technology can assist the assessment of large volumes of financial activity. The Financial Action Task Force (FATF, 2021) examined opportunities for advanced analytics alongside challenges involving data quality, governance and responsible implementation [3]. This institutional review supported investigation of technology-assisted monitoring, but it did not establish that any particular model would be effective in Zimbabwean mobile-money operations. Performance therefore required evaluation within a clearly defined setting.

Jullum et al. (2020) used supervised machine learning on transaction and investigation data from a Norwegian bank. Their study found that a predictive approach could improve the prioritisation of transactions for investigation compared with the bank's existing approach [4]. Its relevance was the use of historical information to support AML decisions. However, evidence from one banking environment did not demonstrate equivalent performance for mobile wallets and cash agents; applying its conclusions to Zimbabwe would require further validation.

Tertychnyi et al. (2022) examined time-aware predictive AML monitoring using customer-history snapshots and an alert policy. Their evaluation considered the timing of detection and

the avoidance of redundant alerts, demonstrating the value of assessing a monitoring process beyond a single classification score [5]. The work informed the present study's attention to chronological behaviour and alert handling. Its institution-specific evaluation, however, did not establish performance for the local mobile-money setting.

Weber et al. (2019) investigated graph-based financial forensics using Bitcoin transaction data. Their comparison included graph convolutional networks and conventional classifiers, with Random Forest performing strongly in the reported experiments [6]. The study demonstrated a way to examine relationships between transactions and cautioned against assuming that a more complex model necessarily performed better. Cryptocurrency transaction graphs did not represent cash-agent operations directly, but the relational perspective was relevant to wallet-to-wallet monitoring.

Altman et al. (2023) developed a synthetic financial-transaction generator and AML datasets to support reproducible model comparisons where shareable operational data were scarce [7]. Their contribution provided a controlled basis for investigating transaction patterns. Nevertheless, synthetic benchmarks depended on their simulation assumptions and did not establish effectiveness on confidential operator data. This distinction was relevant to both behavioural evaluation and the interpretation of model results in an academic prototype.

Within Sub-Saharan Africa, mobile money combined wallet transfers with the activities of cash-in and cash-out agents. The GSMA's 2025 industry report documented the region's central role in the mobile-money sector [1]. These service characteristics made the relationship between customer histories, wallet connections and agent activity relevant to monitoring. Regional adoption established the importance of the application area; it did not, by itself, establish the prevalence of money laundering.

Whisker and Lokanan (2019) reviewed mobile-money characteristics and analysed their implications for AML and counter-terrorist-financing controls. Their thematic analysis identified threats associated with mobile-money services and discussed measures for addressing them [8]. It helped establish why the monitoring of mobile-money channels, including agent-mediated activity, deserved attention. However, it did not provide an experimentally validated agent-behaviour classifier. The present study therefore treated agent activity as a behavioural monitoring dimension whose calculations and predictive usefulness required explicit evaluation.

In Zimbabwe, Chitimira and Torerai (2021) examined mobile-money regulation, technological innovation and financial inclusion through legal and regulatory analysis. They highlighted the

relationship between the regulation of mobile-money providers and inclusive financial services [9]. Their work established local institutional relevance but did not evaluate a machine-learning transaction-monitoring system. The national financial-inclusion strategy also provided a policy context for examining usable and accountable financial services [2].

Taken together, these studies established the potential of predictive monitoring, historical analysis and transaction relationships, while also identifying data and contextual limitations. The specific gap addressed here was the integration and demonstrable evaluation of those ideas within a local mobile-money prototype that connected transaction integrity, structuring, wallet/network and agent behaviour to an authorised investigation process. The reviewed evidence did not establish how that complete workflow would perform in this setting. The study addressed this bounded implementation and evaluation gap; it did not claim that AML monitoring research or mobile-money controls were absent in Zimbabwe.

1.3 Problem Definition

Suspicious mobile-money activity can be distributed across repeated transactions, connected wallets and agent interactions. Reviewing isolated amounts can obscure these relationships, while repetitive alerts can increase investigators' workloads [4], [5], [8]. The resulting challenge is to assess historical behaviour and prioritise relevant cases without losing transaction context. In the Zimbabwean mobile-money setting, the reviewed evidence did not establish the effectiveness of an integrated local monitoring and investigation workflow. This limits the basis for assessing its practical usefulness.

1.4 Aim

To develop a mobile-money AML decision-support system for the Zimbabwean context that integrates transaction processing, historical behavioural analysis, machine-learning classification and authorised investigation of suspicious activity.

1.5 Objectives

The study was organised around five process-based objectives within a 12-week development schedule. The measurable criteria below define acceptance requirements; their inclusion does not imply that every test has already passed. Test procedures and achieved results belong in Chapters 3 and 4.

1. To implement mobile-money transaction processing by the end of Week 3, with correct balance reconciliation and single processing of repeated requests in all 12 predefined tests covering Cash-In, Cash-Out and Wallet-to-Wallet transactions, including valid payments, invalid requests, insufficient funds and duplicate requests.

2. To implement historical behavioural analysis by the end of Week 6 and verify its calculations against independently calculated expected values for six controlled histories: one routine and one suspicious-pattern history for each of structuring, wallet/network and agent behaviour, with no use of future transaction information.

3. To evaluate four candidate classification algorithms and integrate one primary classifier by the end of Week 9, documenting a confusion matrix, precision, recall, macro-F1, false-positive rate and training duration for each candidate under the same development-validation protocol, and verifying that stored decisions reproduce the selected model's outputs.

4. To implement an authorised alert-review process by the end of Week 10, passing six predefined tests covering AI-only, rule-only and combined alerts, authorised review and status recording, denial of customer access to compliance cases, and delivery of notifications only to authorised staff.

5. To implement reliable post-commit monitoring by the end of Week 12, passing five predefined tests for fresh-history verification, concurrent-history change, inference failure, audited inference retry and repeated payment requests, with no duplicated financial transaction or unexplained balance change in any test.

1.5.1 Algorithm-Specific SMART Commitments

The four candidate-algorithm commitments operationalise Objective 3 and are not additional research objectives. Each required one documented development-validation result using the same data partition and evaluation measures, together with recorded training time in seconds. Completion deadlines were expressed relative to the start of the 12-week schedule. Model selection was reserved for the comparative evaluation, rather than assumed in this chapter.

Logistic Regression: To establish and evaluate a linear classification baseline by the end of Week 7, producing the confusion matrix and four classification measures specified in Objective 3, together with a recorded training duration for comparison with the other candidates.

Random Forest: To evaluate an ensemble of decision trees by the end of Week 8, producing the same result set and a recorded training duration under the shared protocol, and comparing its errors with those of the linear baseline.

Histogram-based Gradient Boosting: To evaluate a boosted-tree candidate by the end of Week 8, producing the same result set and a recorded training duration, and documenting its false-positive and missed-pattern behaviour relative to the other candidates.

XGBoost: To evaluate a second boosted-tree implementation by the end of Week 9, producing the same result set and a recorded training duration, and completing a four-candidate comparison that supports a documented primary-model selection.

For these commitments, the four classification measures were precision, recall, macro-F1 and false-positive rate, supported by the confusion matrix; training duration was recorded separately. Measurement of classification quality was required even when a candidate performed poorly. No fixed accuracy percentage or unmeasured execution-speed guarantee was treated as an achieved result.

1.6 Limitations

Access to operational data was limited. Confidential mobile-money transaction records and confirmed investigation outcomes were unavailable to the project, so synthetic records were used. Such records allowed controlled testing but reflected the assumptions used to create them and could omit legitimate behaviour, rare suspicious patterns and changes in real customer activity.

Model evaluation did not establish operational generalisation. Available comparative results were obtained from synthetic development-validation data. They were not an independent external test or a demonstration of effectiveness on a mobile-money operator's network. Reported classification measures therefore had to be interpreted within that restricted evaluation setting.

The classifier could miss suspicious patterns and could flag routine activity. Aggregate scores could also conceal differences between structuring, wallet/network and agent scenarios. Historical-data quality and coverage affected the behavioural information available for a decision. These limitations required separate reporting of errors and explicit handling of unavailable inference results.

Experiments were conducted on one local computer using the hardware described in Section 1.8. Recorded training times reflected that environment and the specific experiment

configuration. They did not measure concurrent-user capacity, live transaction latency or operator-scale throughput. The study did not establish how performance would change across other hardware or larger operational workloads; it did not attribute unrecorded experimental restrictions to computing resources.

1.7 Delimitations

The study used Zimbabwean mobile-money services as its application context and implemented a local academic prototype. It did not connect to a live operator network, telecommunications infrastructure or external regulatory reporting system. Reports and investigation records produced within the prototype were internal records.

The transaction scope covered Cash-In, Cash-Out and Wallet-to-Wallet operations. The behavioural scope covered structuring, wallet/transaction-network relationships and agent activity. The classification task distinguished routine from suspicious-pattern behaviour, while independent AML rules remained a separate source of alerts.

The system supported authorised transaction monitoring and case review. It did not replace customer-identification obligations, investigative judgement or legal determinations. Biometric identification, cryptocurrency analysis, live SIM-registration checks, device intelligence and geolocation-based detection were outside the research scope. The study did not claim that its simulation represented all Zimbabwean providers or regulatory jurisdictions.

1.8 Development Instruments

Development instruments supported application programming, relational data storage, behavioural analysis, model evaluation and local testing. The versions below identify the inspected local environment; they do not imply that all software components ran within one Python environment.

1.8.1 Hardware Instruments

Development was performed on a computer with an Intel Core i5-9400H processor rated at 2.50 GHz and approximately 15.73 GiB of operating-system-visible memory, corresponding to the nominal 16 GB memory specification. The machine hosted the local application, database and model-development tools. These specifications provided context for the recorded experiment timings.

1.8.2 Software Instruments

Table 1.1: Inspected software instruments and versions

Instrument / version	Role
Windows 11, build 26300.9550	Local development operating system.
Python 3.14.3	Application programming and model execution.
Flask 3.1.3	Web routes, request handling and application services.
Flask-SocketIO 5.6.1	Server-side real-time event communication.
Socket.IO browser client 4.7.5	Browser-side event delivery.
React 18.3.1	Interactive browser components.
MySQL Server 8.0.46	Relational storage and transaction management.
mysql-connector-python 9.7.0	Application connection to MySQL.
scikit-learn 1.9.0	Classification, preprocessing and evaluation utilities.
XGBoost 3.4.1	Boosted-tree experiments and the separate model-runtime environment.
NumPy 2.5.0 (application); 2.4.3 (model runtime)	Numerical arrays in the respective environments.
pandas 3.0.3 (application)	Tabular data manipulation.
SciPy 1.18.0; joblib 1.5.3 (model runtime)	Scientific computation and model serialisation.
Git 2.53.0.windows.2	Source-code version control.
HTML, CSS and JavaScript	Browser structure, presentation and interaction; web languages rather than separately versioned Python packages.

1.9 Work Plan

The retained work plan covered 12 relative development weeks. Table 1.2 identifies activities, deadlines and evidence, while Figure 1.1 shows their planned overlap. The schedule defines development milestones rather than retrospectively certifying completion dates. Documentation ran throughout the project, and final verification assessed the five objectives together.

Table 1.2: Twelve-week work plan

Activity	Weeks	Required evidence
Requirements and design	1–2	Defined scope, workflows and acceptance-test matrix.
Transaction processing	2–3	Objective 1: transaction and balance checks.
Controlled histories and behaviour	3–6	Objective 2: six independently checked history fixtures.
Logistic Regression baseline	7	Baseline metrics and training duration.
Random Forest and histogram boosting	7–8	Comparable candidate metrics and training durations.
XGBoost and model comparison	8–9	Objective 3: four-candidate comparison and primary-model integration.
Alert review and access control	9–10	Objective 4: six alert and authorisation checks.
Reliable post-commit processing	10–12	Objective 5: five integrity and recovery checks.
Documentation and demonstration	1–12	Dissertation, evidence register and final demonstration.

Activity	1	2	3	4	5	6	7	8	9	10	11	12
Requirements	■	■										
Transactions		■	■									
Behaviour			■	■	■	■						
LR baseline							■					
RF and HGB							■	■				
XGB / comparison								■	■			
Alert review									■	■		
Reliable monitoring										■	■	■
Documentation	■	■	■	■	■	■	■	■	■	■	■	■

Figure 1.1: Twelve-week project Gantt chart; shaded markers indicate planned activity.

1.10 Justification / Rationale

The project was justified by the importance of maintaining useful financial services alongside accountable monitoring. Mobile money's regional role and Zimbabwe's financial-inclusion priorities made this a relevant application area [1], [2]. Investigators required a way to examine related activity and distinguish the basis of an alert before deciding whether further review was warranted.

Historical and relational analysis offered a practical basis for examining activity that an isolated transaction view might overlook. Previous predictive-monitoring studies supported investigating machine learning as an aid to prioritisation, while also showing why evaluation needed to account for errors and the alert workflow [4], [5]. A local prototype provided a bounded setting in which those functions could be demonstrated and questioned.

The contribution was an integrated, testable mobile-money decision-support workflow connecting transaction integrity, behavioural analysis, classification and authorised investigation. Its academic value lay in evaluating how these processes worked together and documenting their limitations. Its potential practical value was to help investigators trace alerts to their recorded basis while preserving financial and audit records. The project did not claim a new learning algorithm, proven crime reduction or operational effectiveness beyond the evidence collected.

1.11 Summary

This chapter established the difficulty of assessing suspicious activity across mobile-money histories and relationships. It defined an overall development aim and five measurable process objectives within a 12-week schedule. It also stated the study's local scope, data and evaluation limitations, development instruments and justification. The proposed contribution concerned an integrated and demonstrable investigation-support workflow. Chapter 2 reviews the relevant literature and existing systems in greater depth, providing the basis for the design and evaluation choices developed in later chapters.

1.12 References

[1] GSMA, The State of the Industry Report on Mobile Money 2025, 2025. [Online]. Available: https://www.gsma.com/sotir/wp-content/uploads/2025/04/The-State-of-the-Industry-Report-2025_English.pdf. Accessed: Oct. 14, 2026.

[2] Reserve Bank of Zimbabwe, Zimbabwe National Financial Inclusion Strategy II, 2022–2026, 2022. [Online]. Available: https://www.rbz.co.zw/documents/BLSS/FinancialInclusion/Zimbabwe_National_Financial_Inclusion_Strategy_II_2022-2026.pdf. Accessed: Oct. 14, 2026.

[3] Financial Action Task Force, Opportunities and Challenges of New Technologies for AML/CFT, Paris, France, 2021. [Online]. Available: https://www.fatf-gafi.org/content/dam/fatf-gafi/guidance/Opportunities-Challenges-of-New-Technologies-for-AML-CFT.pdf. Accessed: Oct. 14, 2026.

[4] M. Jullum, A. Løland, R. B. Huseby, G. Ånonsen, and J. Lorentzen, "Detecting money laundering transactions with machine learning," Journal of Money Laundering Control, vol. 23, no. 1, pp. 173–186, 2020, doi: 10.1108/JMLC-07-2019-0055.

[5] P. Tertychnyi, M. Godgildieva, M. Dumas, and M. Ollikainen, "Time-aware and interpretable predictive monitoring system for Anti-Money Laundering," Machine Learning with Applications, vol. 8, Art. no. 100306, 2022, doi: 10.1016/j.mlwa.2022.100306.

[6] M. Weber et al., "Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics," KDD Workshop on Anomaly Detection in Finance, 2019. [Online]. Available: https://arxiv.org/abs/1908.02591. Accessed: Oct. 14, 2026.

[7] E. Altman, J. Blanuša, L. von Niederhäusern, B. Egressy, A. Anghel, and K. Atasu, "Realistic Synthetic Financial Transactions for Anti-Money Laundering Models," Advances in Neural Information Processing Systems, vol. 36, 2023. [Online]. Available: https://arxiv.org/abs/2306.16424. Accessed: Oct. 14, 2026.

[8] J. Whisker and M. E. Lokanan, "Anti-money laundering and counter-terrorist financing threats posed by mobile money," Journal of Money Laundering Control, vol. 22, no. 1, pp. 158–172, 2019, doi: 10.1108/JMLC-10-2017-0061.

[9] H. Chitimira and E. Torerai, "The Nexus between Mobile Money Regulation, Innovative Technology and the Promotion of Financial Inclusion in Zimbabwe," Potchefstroom Electronic Law Journal, vol. 24, 2021, doi: 10.17159/1727-3781/2021/v24i0a10739.